

Analysis Report: Digit Reversal Without an Apology

Name: Francis Reuben R

USN: 1RUA25CSE0168 (RVUN25CSE408)

Roll Number: 51

Course: English

Date: December 1, 2025

Report

In my presentation, I talked about digit reversal and palindromic numbers, which turned out to be way more interesting than I thought when I first picked this topic. I started by explaining what palindromes are, which are basically numbers that read the same forwards and backwards, like 121. The concept seems simple at first, but it opens up a lot of questions about how numbers behave.

Then I got into the Lychrel problem, which focuses on the number 196. What got me about this is that nobody knows if 196 ever becomes a palindrome, even after running it through computers millions of times. It's crazy to think that something so simple can stump mathematicians for decades. I also showed how these patterns work in different number systems, not just the decimal one we use every day. Binary and hexadecimal systems have their own palindromes too.

I looked at how reversing digits and adding them creates sequences that sometimes turn into palindromes and sometimes just keep going forever. What I found through making this presentation is that simple math operations can lead to problems that mathematicians still can't figure out. This whole topic made me think differently about numbers and showed me that there's a lot we don't understand about basic arithmetic, which honestly makes math more interesting to me now.

References

Pudwell, L. (2006). Digit reversal without apology. *arXiv preprint arXiv:math/0511366v2*.

Reuben R, F. (2025). *Digit reversal without an apology* [Presentation slides].